ALL INDIA TEST SERIES

From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

CONCEPT RECAPITULATION TEST - III

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

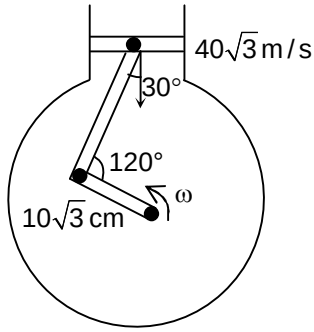
SECTION – A

Single Correct Choice Type

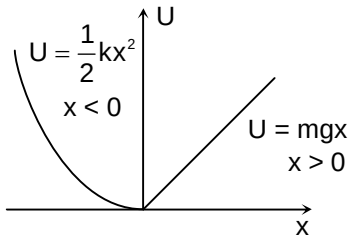
This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. An electron is released from rest at one point in a uniform electric field and travels a distance of 10 cm in 10^{-7} seconds. Potential difference across the points is
(A) 11.375 volts (B) 10 volts
(C) 5 volts (D) 5.7 volts

2. A snapshot of a petrol engine is given in which piston is moving downwards with velocity $40\sqrt{3}\,m/s$. Find the angular velocity of the shaft.
(A) 400 rad/s
(B) 300 rad/s
(C) 200 rad/s
(D) 500 rad/s

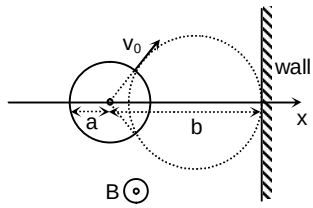


3. A particle of mass m moves in the potential shown in the figure. The particle has energy E. The period of the motion the particle is
(A) $2\pi\sqrt{\frac{m}{k}}+2\sqrt{\frac{2E}{mg^2}}$
(B) $2\pi\sqrt{\frac{m}{k}}+\sqrt{\frac{2E}{mg^2}}$
(C) $\pi\sqrt{\frac{m}{k}}+2\sqrt{\frac{2E}{mg^2}}$
(D) $\pi\sqrt{\frac{m}{k}}+\sqrt{\frac{2E}{mg^2}}$



Space for Rough work

4. A electron is projected normally from the surface of a sphere with speed v_0 in a uniform magnetic field perpendicular to the plane of the paper such that its strikes symmetrically opposite on the sphere with respect to the x-axis. Radius of the sphere is 'a' and the distance of its centre from the wall is 'b'. What should be magnetic field such that the charge particle just escapes the wall



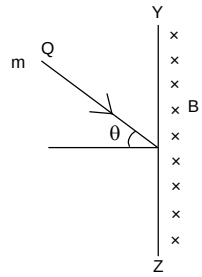
(A) $B = \dfrac{2bmv_0}{(b^2 - a^2)e}$

(B) $B = \dfrac{2bmv_0}{(a^2 - b^2)e}$

(C) $B = \dfrac{(a^2 - b^2)}{2bmv_0}$

(D) $B = \dfrac{me(b^2 - a^2)}{2bmv_0}$

5. A particle with positive charge Q and mass m enters a magnetic field of magnitude B, existing only to the right of boundary YZ. The direction of the motion of the particle is perpendicular to the direction of B. Let $T = 2\pi\left(\dfrac{m}{QB}\right)$. The time spent by the particle in the field will be



(A) $T\theta$

(B) $2T\theta$

(C) $T\left(\dfrac{\pi + 2\theta}{2\pi}\right)$

(D) $T\left(\dfrac{\pi - 2\theta}{2\pi}\right)$

6. If net external torque, about a point, acting on the system is zero, then we can surely say

(A) Kinetic energy of the system remains constant

(B) Mechanical energy of the system remains constant

(C) Torque of Internal forces is zero

(D) Momentum of system will remain constant.

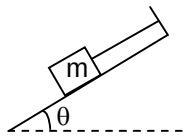
7. A horizontal spring mass system of mass M performs simple harmonic motion of amplitude a_0 and time period T_0. When the mass M passes through mean position, another stationary mass M sticks to it and both move together. If a and T be new amplitude and time period, then

(A) $a = \dfrac{a_0}{\sqrt{2}}$ (B) $a = a_0\sqrt{2}$

(C) $T = \dfrac{T_0}{\sqrt{2}}$ (D) None

Space for Rough work

8. A metal block of mass m is placed on a smooth metallic plane in support with string as shown in diagram. If, after long time due to corrosion, the contact surface becomes rough with coefficient of friction μ, then friction force acting on the block will be



(A) $\mu mg\cos\theta$

(B) $mg\sin\theta$

(C) mg

(D) None

9. A uniform rod of mass 2M and length 2L is bent at its mid point to form an L shape. Then, moment of inertia of the system about an axis symmetrical to both straight parts of L shaped rod is

(A) $\frac{7}{5}ML^2$ (B) $\frac{1}{3}ML^2$

(C) $\frac{7}{24}ML^2$ (D) $\frac{ML^2}{2}$

10. Which of the following is correct about principle of conservation of momentum?

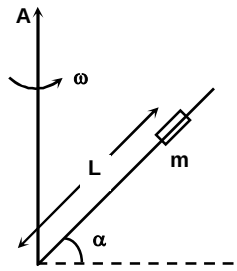
(A) Conservation of momentum can be applied only in absence of external forces

(B) Conservation of momentum can be applied only during collisions of bodies

(C) Conservation of momentum can be applied in a process even in the presence of external forces

(D) Conservation of momentum is not applicable in rocket propulsion

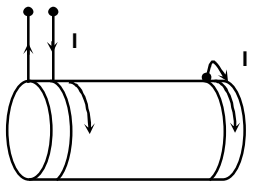
11. A rod revolves with a constant angular velocity ω about a vertical axis A. The rod makes a fixed angle α with the horizontal. A body of mass m can slide along the rod and it has tendency to slide down, the angle of friction being β. The value of α so that "m" remains at rest (relative to the rod) is given by



(A) $\tan(\alpha-\beta)=\frac{\omega^2 L}{g}\cos\alpha$ (B) $\tan(\alpha+\beta)=\frac{\omega^2 L}{g}\cos\alpha$

(C) $\omega^2 L\sin(\alpha+\beta) = g\cos\beta$ (D) $\omega^2 L\sin(\alpha-\beta) = g\cos\alpha$

12. Two long cylindrical coils with uniform winding of the same length and nearly the same radius having inductances L_1 and L_2. The coils are coaxially inserted into each other and connected to a current source as shown in the figure. The inductance L of the composite system is



(A) $L_1 + L_2$

(B) $L_1L_2(L_1+L_2)$

(C) $L_1 + L_2 + \sqrt{L_1L_2}$

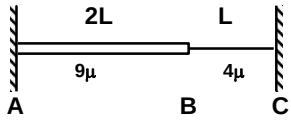
(D) $L_1 + L_2 + 2\sqrt{L_1L_2}$

Space for Rough work

13. A disturbance can be written as $y(x, t) = \left[e^{-(x/b)^2} e^{2xt/b} e^{-t^2}\right]^a$. The disturbance is
(A) not a travelling wave
(B) a travelling wave with speed v = a
(C) a travelling wave with speed v = a/b
(D) a travelling wave with speed v = b

14. Which of the following is cause of dispersion of a white light by a prism
(A) waves of different frequencies lose energy at different rates.
(B) waves of different amplitudes lose energy at different rate.
(C) waves of different frequencies travel through the medium at different speeds
(D) waves of different amplitudes travels through the medium with different wavelengths.

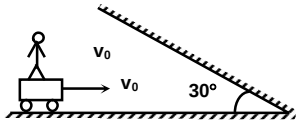
15. A thick rope of length 2L and linear mass density 9μ is joined at B to a thin rope of length L and linear mass density 4μ. The system is horizontally stretched by the two vertical wall A and C. Assuming B to be a node, find the minimum number of loops in the thick rope.



(A) 1 (B) 2
(C) 3 (D) 4

16. In an L-C-R series circuit, it is found that the power factor is maximum. Then the angular frequency of the emf source must be equal to
(A) $\frac{1}{\sqrt{LC}}$ (B) zero
(C) a very large number (D) none of these

17. A car approaches an inclined wall (see in the figure) with speed v_0 an a horizontal road. When the horn of frequencies f_0 is sounded, the beat frequencies heard by the person on the car is (v = speed of sound)



(A) $\frac{v_0 f_0}{v - v_0}$ (B) $\frac{2v_0 f_0}{2v - v_0}$
(C) $\frac{v_0 f_0}{2(v - v_0)}$ (D) $\frac{2v_0 f_0}{v - v_0}$

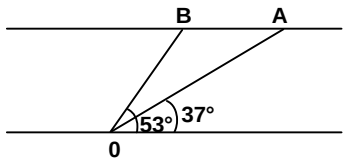
18. The speed of the colliding electron is increased in an X-ray tube, then
(A) the cutoff wavelength must increase
(B) the cutoff frequency must increase
(C) the intensity of the X-rays must increase.
(D) the types of characteristic X-rays must increase

Space for Rough work

19. If white light is used in an YDSE setup, the nearest bright fringe from the central fringe will be
(A) white (B) violet
(C) red (D) yellow

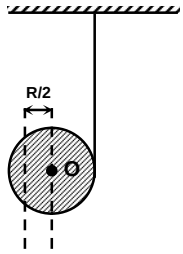
20. A particle is placed in a cart. Initial location of the particle is taken as origin in ground frame. At t = 0, the cart starts moving along x-axis with uniform velocity 2m/s and at the same instant the particle starts moving with $2x\hat{j}$ relative to cart where x is x-coordinate of the cart. Which of the following is the radius of curvature of the trajectory of the particle in ground frame when it's tangential acceleration is equal to normal acceleration in magnitude?
(A) $\frac{1}{\sqrt{2}}$m (B) $\frac{2}{\sqrt{2}}$m
(C) $\frac{3}{2\sqrt{2}}$m (D) $\frac{4}{\sqrt{2}}$m

21. A boat crosses a 2.4km wide river to reach the a point A on the opposite bank, located at 37° with the flow, in 500s. The same boat takes the same time (500s) to reach another point B on the opposite bank located at 53° with the flow. the speed of flow of the river is



(A) 5 m/s
(B) 10 m/s
(C) 15 m/s
(D) none of these

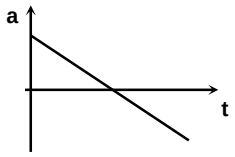
22. A disc of mass M and radius R is wounded by massless inextensible thread. One end of string is tied to ceiling and system is released. Then which of the followings are/is false.

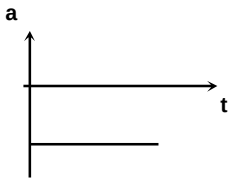


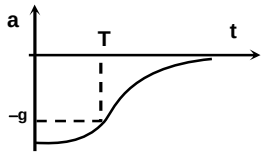
(A) Tension in the thread mg/3
(B) acceleration of disc 2g/3
(C) angular momentum of disc is conserved about point of contact on rim with thread.
(D) angular momentum is conserved about any point on vertical line at distance x = R/2 from centre toward left.

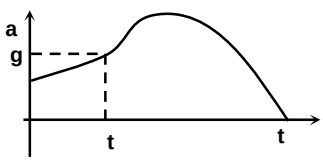
Space for Rough work

23. A ball is thrown vertically upward at time t = 0. Viscous force of air resistance is not negligible and the acceleration due to gravity is g. The ball reaches maximum height at t = T and then descends attaining terminal speed. Which graph shows the best variation with time t of the acceleration a of the ball.

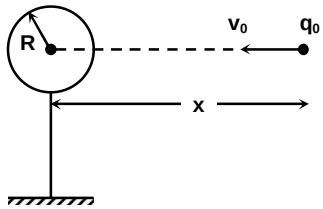
(A) 

(B) 

(C) 

(D) 

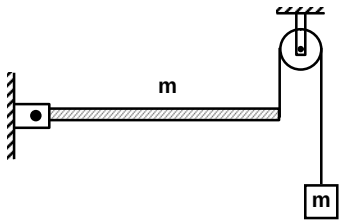
24. An uncharged sphere of radius R is connected by zero resistance wire to earth. A charged particle q_0 is moving towards centre of sphere with constant velocity v_0. Find current flowing in wire. When charged particle is at distance x. q_0 = 10 C, R = 1 m, v_0 = 10 m/s, x = 2 m.



(A) 25 A (B) 5A

(C) infinity (D) zero.

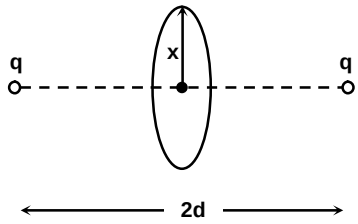
25. A rod and a block are of same mass. Initially rod is in horizontal position. What will be acceleration of tip of the rod. When system is released from this position



(A) zero (B) $\frac{3g}{4}$

(C) $\frac{3g}{8}$ (D) $\frac{3g}{2}$

26. Two identical charges q are separated by distance 2d and fixed at this position. Third charge $-q_0$ is revolving in a circle of radius x in plane perpendicular to line joining charges and centre coincides with mid point of line joining charges. The value of x for which speed of particle is maximum is [Consider gravity free space.]

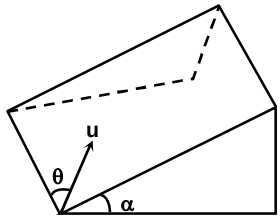


(A) $\frac{d}{\sqrt{2}}$ (B) $\sqrt{2}$ d

(C) d (D) 2d

Space for Rough work

27. A particle is thrown at an angle θ with horizontal along the inclined plane of inclination α. Find the maximum height from the ground reached by particle



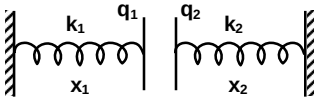
(A) $\frac{u^2 \sin^2 \theta}{2g}$ (B) $\frac{u^2 \sin^2 \theta \sin \alpha}{2g}$

(C) $\frac{u^2 \sin^2 \theta}{2g \sin \alpha}$ (D) $\frac{u^2 \sin^2 \theta \sin^2 \alpha}{2g}$

28. A charged particle of charge q is revolving in a circle of radius R of time period T = 9 sec. A point P at distance 2R from centre of circle in same plane record magnetic field. Find minimum time interval between two successive times at which magnetic field is zero at that point

(A) 9 sec
(B) 4.5 sec
(C) 3 sec
(D) magnetic field at point P will always have non-zero value.

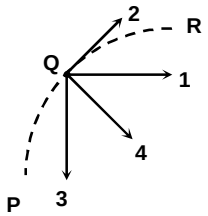
29. Two plates of capacitors are provided charges q_1 and q_2 and connected with springs of constant k_1 and k_2 as shown in the diagram. Find the ratio of elongation in two springs (x_1/x_2). (springs are of dielectric material]



(A) $\frac{k_2}{k_1}$ (B) $\left(\frac{k_2}{k_1}\right)^2$

(C) 1 (D) $\left(\frac{k_1}{k_2}\right)^2$

30. A body moves along a path PQR from P to R shown as a dashed line in figure. When the particle is at Q, its speed is decreasing. The acceleration of the particle at Q is best represented by the



(A) 1 (B) 2
(C) 3 (D) 4

Space for Rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

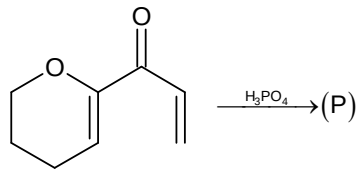
This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. The correct order of leaving group for aromatic nucleophilic substitution is:

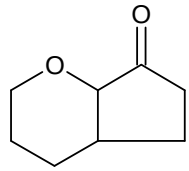
(A) $F^- > Cl^- > I^- > Br^- > NO_2^-$ (B) $F^- > NO_2^- > Cl^- > Br^- > I^-$

(C) $NO_2^- > Cl^- > Br^- > I^- > F^-$ (D) $I^- > Br^- > Cl^- > F^- > NO_2$

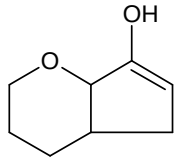
2.



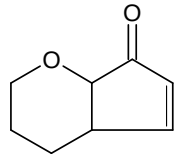
(A)



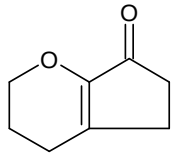
(B)



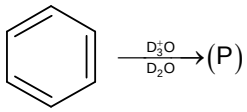
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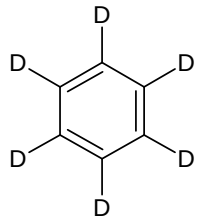
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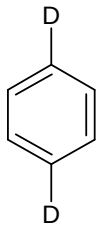


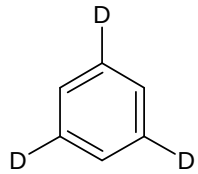
Rough Work

3. 

The product (P) formed finally is?

(A) 

(B) 

(C) 

(D) None of these

4. 'Talc' is one of the very soft material and it is used in talcum powder having molecular formula $Mg_3Si_4O_{10}(OH)_2$. What type of silicate it is?
(A) Two dimensional sheet silicate
(B) Three dimensional sheet silicate
(C) Cyclic silicate
(D) Chain silicate

5. What is the false about H_2O_2?
(A) Acts both as oxidising and reducing agent.
(B) It is an antiseptic and germicide for working wounds, teeth and ears under the name perhydrol.
(C) H_2O_2 is prepared by adding ice cold dilute H_2SO_4(20%) on Na_2O_2.
(D) Two OH bonds of H_2O_2 lie in the same plane.

6. Hydrazine may be obtained from the reaction between ammonia and hydrogen peroxide:

$2NH_3(g) + H_2O_2(\ell) \longrightarrow N_2H_4(\ell) \qquad \Delta_r H^o = -241.0\ kJ/mol.$

What is the standard enthalpy change per mole for the decomposition of hydrazine to its elements into their standard state? Given the standard enthalpy changes of formation (in kJ/mol) are:

$NH_3(g)$ = - 46.1

$H_2O_2(\ell)$ = - 187.8

$H_2O\ (\ell)$ = - 285.8

(A) - 50.6 kJ/mol
(B) + 50.6 kJ/mol
(C) - 25.3 kJ/mol
(D) +25.3 kJ/mol

Rough Work

7. $(A) + X^- + K_2Cr_2O_7(s) + H_2SO_4(conc.) \xrightarrow{\Delta} \underset{(\text{Red vapour})}{B} + \underset{(\text{Reddish brown vapour})}{C(g)}$

$C + H_2O \longrightarrow$ Brown red colouration $\xrightarrow{NaOH} (E) + (F)$

colourless solution

$B + NaOH \longrightarrow (D)$

Yellow solution

$(D) + Pb(OAc)_2 \xrightarrow{CH_3COOH}$ Yellow solid

What is (D) in the above sequence?

(A) Na_2CrO_4 (B) $NaBrO_3$

(C) NOBr (D) None of these

8. H_2S is bubbled into 0.2 M NaCN solution which is 0.02 M in each $[Cd(CN)_4]^{2-}$ and $[Ag(CN)_2]^-$. The H_2S produces 1×10^{-9} M sulphide ion in the solution.

Given: $K_{sp}Ag_2S = 1\times10^{-50}\ M^3; K_{sp}CdS = 7.1\times10^{-28}\ M^2$

$K_{instability}\left[Ag(CN)_2\right]^{-1} = 1\times10^{-20}\ M^2,\ K_{instability}\left[Cd(CN)_4\right]^{2-} = 7.8\times10^{-18}\ M$

Identify the correct statement:

(A) Ag_2S precipitates first from the solution.

(B) CdS precipitates first from the solution.

(C) None of them precipitates under the given conditions.

(D) Ag_2S precipitates at a sulphide concentration $1 \times 10^{+15}$ M.

9. Predict coordination number of the cation in the crystals of the following compounds:

1. MgO : r_c = 0.65 Å ; r_a = 1.40 Å

2. MgS : r_c = 0.65 Å ; r_a = 1.84 Å

(Where r_c and r_a represents the radius of cation and anion respectively).

(A) 6, 4 (B) 4, 6

(C) 3, 4 (D) 6, 8

10. The chlorate ion can disproportionate in basic solution according to reaction,

$2ClO_3^- \rightleftharpoons ClO_2^- + ClO_4^-$

What is the equilibrium concentration of perchlorate ions from a solution initially at 0.1 M in chlorate ions at 298 K?

Given: $E^o_{ClO_4^-/ClO_3^-} = 0.36$ V and $E^o_{ClO_3^-/ClO_2^-} = 0.33$ V at 298 K.

(A) 0.019 M (B) 0.024 M

(C) 0.1 M (D) 0.19 M

Rough Work

11. A d-Block element forms octahedral complex but its magnetic moment remains same either in strong field or in weak field ligand. Which of the following is/are correct?
(A) Element always forms colourless compound.
(B) Number of electrons in t_{2g} orbitals are higher than in e_g orbitals.
(C) It can have either d^3 or d^8 configuration.
(D) It can have either d^7 or d^8 configuration.

12. The Schrodinger wave equation for hydrogen atom is :

$$\psi(r)_{n,\ell} = \frac{1}{16\sqrt{4}}\left(\frac{1}{a_0}\right)^{3/2}\left[(\sigma-1)(\sigma^2-8\sigma+12)\right]e^{-\sigma/2}$$

Where a_0 and z are constant in which answer can be expressed and $\sigma = \frac{2r}{a_0}$. Then the minimum and maximum position of radial nodes from nucleus respectively are:

(A) a_0, $3a_0$ (B) $\frac{a_0}{2}, a_0$

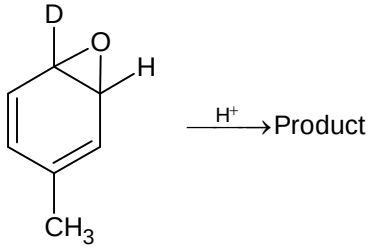
(C) $\frac{a_0}{2}, 3a_0$ ` (D) $\frac{a_0}{2}, 4a_0$

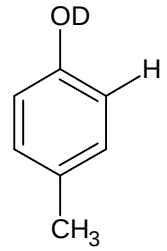
13. Equivalent weight of H_3PO_2, when it disproportionate into PH_3 and H_3PO_3 is:
(Given M is the molecular mass of H_3PO_2)
(A) M (B) M/2
(C) M/4 (D) 3M/4

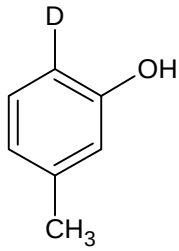
14. A complex of iron and cyanide ions is 100% ionized at 1 molal concentration. If its elevation in boiling point is 2.08^o, then the complex is (Given $K_b = 0.52^o$ C mol^{-1} kg)
(A) $Fe_4[Fe(CN)_6]_3$ (B) $K_3[Fe(CN)_6]$
(C) $K_4[Fe(CN)_6]$ (D) $Fe_3[Fe(CN)_6]_2$

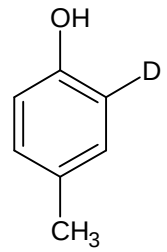
Rough Work

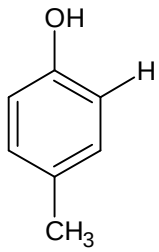
15. What will the major product of the following reaction?

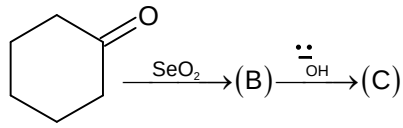


(A) 

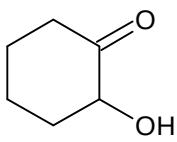
(B) 

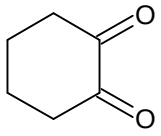
(C) 

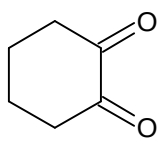
(D) 

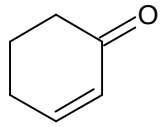
16. 

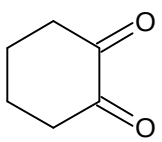
Compounds (B) and (C), respectively are:

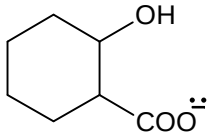
(A) 

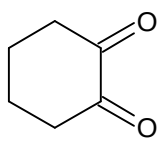
and 

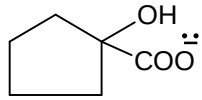
(B) 

and 

(C) 

and 

(D) 

and 

17. In the commercial electrochemical process for aluminium extraction, the electrolyte used is:
(A) $Al(OH)_3$ in NaOH solution
(B) An aqueous solution of $Al_2(SO_4)_3$
(C) A molten mixture of Al_2O_3 and Na_3AlF_6
(D) A molten mixture of Al_2O_3 and $Al(OH)_3$

Rough Work

18. DDT is:
(A) antibiotic
(B) a fertilizer
(C) a non-degradable pollutant
(D) dichlorodifluorotitanium

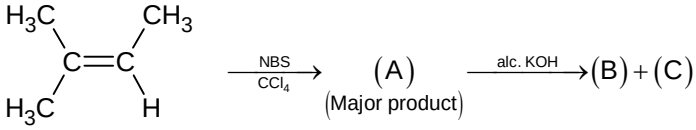
19. Which of the following colloidal solution is used in photography?
(A) Colloidal silver bromide
(B) Colloidal gold
(C) Colloidal silver
(D) Colloidal sulphur

20. The correct increasing order of thermal stabilities of
$\underset{(I)}{K_2CO_3}$ $\underset{(II)}{MgCO_3}$ $\underset{(III)}{CaCO_3}$ $\underset{(IV)}{BeCO_3}$
(A) I < II < III < IV
(B) IV < II < III < I
(C) IV < II < I < III
(D) IV < II < I < III

21. Which of the following molecular orbital have 2-nodal planes?
(A) π_{2px}
(B) π_{2py}
(C) π^*_{2px}
(D) σ_{1s}

22. The reaction, $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is favoured in forward direction by:
(A) increasing the volume of the container.
(B) introducing PCl_5 at constant volume.
(C) adding inert gas at constant pressure
(D) all of the above

23. In the given reaction:

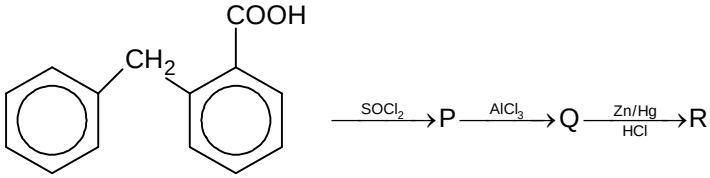


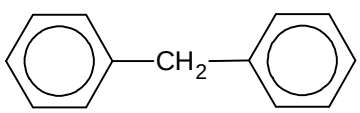
Which of the following is correct?
(A) Compounds (B) and (C) are stereoisomer and both are planar.
(B) Compounds (B) and (C) are not stereoisomer and both are planar.
(C) Compounds (B) and (C) are not stereoisomer and one of them is planar.
(D) Compounds (B) and (C) are stereoisomer and one of them is planar.

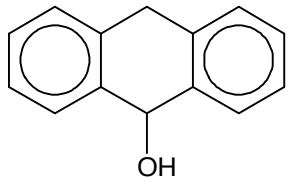
24. For a first order reaction which among the following is incorrect?
(A) $r \propto [\text{conc.}]^1$.
(B) $t_{1/2}$ is independent of initial concentration.
(C) Slope of graph of log(conc.) versus time in $-\frac{K}{2.303}$.
(D) $t_{1/2} \propto [\text{conc.}]^{1/2}$.

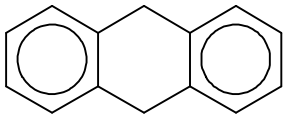
Rough Work

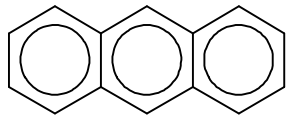
25. What is R in the following set of reactions?



(A) 

(B) 

(C) 

(D) 

26. Which of the following is not correct?

(A) $FeCl_3 + KCNS \longrightarrow$ Red colour (B) $KI + PbCl_2 \longrightarrow$ Yellow ppt.

(C) $NH_4Cl +$ Nesseler's reagent $\longrightarrow$ Blue colour (D) $DMG + Ni^{2+} + NH_4OH \longrightarrow$ Scarlet red ppt.

27. Nucleosides contain:

(A) pentose sugar (B) purine pyrimidine
(C) phosphate (D) both (A) and (B)

28. Which of the following is not a cross linked polymer?

(A) Alkyde resin (B) Bakelite
(C) Vulcanised rubber (D) Nylon-6

29. Acetonitrile on reduction gives:

(A) Propanamine (B) Methanamine
(C) Ethanamine (D) None of these

30. The fundamental unit of linear silicones is:

(A) SiO_4^{4-} (B) $(R_2SiO)_n$
(C) $(RSiO_3)_n$ (D) none of these

Rough Work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Consider the circle c : $x^2 + y^2 = 1$ and the line L : y = m(x + 2). If L intersect c at P and Q, then locus of middle point of PQ is

(A) $(x+1)^2 + y^2 = 1$ (B) $x^2 + (y-1)^2 = 1$

(C) $(x-1)^2 + y^2 = 1$ (D) $x^2 + (y+1)^2 = 1$

2. Let f : (1, ∞) → (0, ∞) be a continuous decreasing function with $\lim_{x\to\infty} \frac{f(4x)}{f(8x)} = 1$, then $\lim_{x\to\infty} \frac{f(6x)}{f(8x)}$ is equal to

(A) $\frac{4}{8}$ (B) $\frac{4}{6}$

(C) $\frac{6}{8}$ (D) 1

3. If the equation $ax^2 - 6xy + y^2 + 2gx + 2fy + c = 0$ represents a pair of lines whose slopes are m and m^2, then sum of all possible values of a is

(A) 17 (B) –19

(C) 19 (D) –17

4. Let g : R → R be a differentiable function such that g(2) = –40 and g′(2) = –5 then $\lim_{x\to 0}\left(\frac{g(2-x^2)}{g(2)}\right)^{\frac{4}{x^2}}$ is equal to

(A) e^{32} (B) $\sqrt{e}$

(C) $\frac{1}{\sqrt{e}}$ (D) e^{-5}

Space for rough work

5. Let f : [0, 1] $\to$ R be a differentiable function with f(0) = 0, then $\lim_{n\to\infty} n^2 \int_0^{1/n} f(t)dt$ equals

(A) $\frac{1}{2}f'(0)$ (B) f′(0)

(C) 2f′(0) (D) 0

6. The values of parameter a such that the line $(\log_2(1 + 5a – a^2))$ x– 5y – $(a^2 – 5) = 0$ is a normal to the curve xy = 1, may be in the interval

(A) $(-\infty, 0)$ (B) $(5, \infty)$

(C) (0, 5) (D) $(-\infty, 0) \cup (5, \infty)$

7. A triangle has vertices A, B, C and their respective opposite sides have tangents a, b, and c. Then triangle is inscribed in a circle of radius R. If b = c = 1 and the altitude from A to side BC has length $\sqrt{\frac{2}{3}}$, then R is equal to

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{2}{\sqrt{3}}$

(C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{3}}{2\sqrt{2}}$

8. Let O be the centre of a circle of radius 1 P and Q are the points on the circle such that $\theta = \angle POQ$ is an acute angle and R is a point outside the circle such that OPRQ is a parallelogram. If the area of the part of the parallelogram that is outside the circle is $f(\theta)$, then $\lim_{\theta\to 0} \frac{f(\theta)}{\theta}$ is equal to

(A) $\frac{3}{\pi}$ (B) $\frac{2}{\pi}$

(C) $\frac{1}{2}$ (D) 1

9. If $\sum_{r=1}^{10} r(r-1)\ {}^{10}C_r = k \cdot 2^9$, then k is equal to

(A) 10 (B) 45

(C) 90 (D) 100

Space for rough work

10. If A and B are square matrices of order 3 where |A| = –2 and |B| = 1 then $|(A^{-1})\ \text{adj}(B^{-1})\ \text{adj}(2A^{-1})|$ is equal to
(A) 8 (B) –8
(C) 1 (D) –1

11. P is a point on the line y + 2x = 1 and Q and R are two points on the line 3y + 6x = 6 such that the triangle PQR is an equilateral triangle. Area of this triangle is equal to
(A) $\frac{1}{15}$ (B) $\frac{1}{5\sqrt{3}}$
(C) $\frac{1}{3\sqrt{5}}$ (D) $\frac{1}{2\sqrt{15}}$

12. Let OABC be a tetrahedron (O beams the origin). If position vectors of A, B and C are $\hat{i}$, $\hat{i}+\hat{j}$ and $\hat{j}+\hat{k}$ respectively, then height of the tetrahedron (taking ABC as base) is equal to
(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$
(C) $\frac{1}{2\sqrt{2}}$ (D) $\sqrt{2}$

13. A circle is inscribed in a right triangle ABC, right angled at C. The circle is tangent to the segment AB at D and length of segments AD and DB are 7 and 13 respectively. Area of triangle ABC is equal to
(A) 91 (B) 96
(C) 100 (D) 104

14. The vector equation of a curve in R^3 denoted by $\vec{r} = (t^2+t-2)\hat{i}+(t+2)\hat{j}+(t+3)^2\hat{k}$ denotes
(A) a plane passing through the origin (B) a plane passing through (0, 0, 1)
(C) a line parallel to the vector $\hat{i}+5\hat{j}-\hat{k}$ (D) a pair of lines intersecting at (0, 0, 1)

15. Rajdhani express stops at six intermediate stations between Kota and Mumbai. Five passengers board at Kota. Each passenger can get down at any station till Mumbai. The probability that all five passengers will get down at different stations, is
(A) $\frac{{}^6P_5}{6^5}$ (B) $\frac{{}^6C_5}{6^5}$
(C) $\frac{{}^7P_5}{7^5}$ (D) $\frac{{}^7C_5}{7^5}$

Space for rough work

16. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = 1$ and $\vec{a} \cdot (\vec{b} \times (\vec{a} \times \vec{b})) = 8$. If angle between $\vec{a}$ and $\vec{b}$ is $\cos^{-1}\frac{1}{\sqrt{2}}$, then magnitude of $\vec{b}$ equals

(A) 2 (B) 4
(C) 8 (D) 16

17. An ellipse in the first quadrant is tangent to co–ordinate axes. If one focus is $F_1(3, 7)$ and other focus is $F_2(d, 7)$ then the value of $\sqrt{3d}$ is equal to

(A) 5 (B) 7
(C) 14 (D) 49

18. The minimum value of the function $f(x) = \int_0^x \frac{d\theta}{\cos\theta} + \int_x^{\pi/2} \frac{d\theta}{\sin\theta}$ where $x \in \left(0, \frac{\pi}{2}\right)$, is

(A) $2\ln(\sqrt{2}+1)$ (B) $\ln(2\sqrt{2}+2)$
(C) $\ln(2+\sqrt{3})$ (D) $\ln(3+\sqrt{2})$

19. Let $I(a) = \int_0^a \ln(\cot a + \tan x)\,dx$, where $a \in \left(0, \frac{\pi}{2}\right)$ then I(1) has value equal to

(A) ln (sin 1) (B) –ln (cos 1)
(C) ln (cosec 1) (D) ln (sec 1)

20. $\lim_{x\to 0}\left(2^{x-1} + \frac{1}{2}\right)^{1/x}$ is equal to

(A) $\sqrt{2}$ (B) $\frac{1}{2}\ln 2$
(C) ln 2 (D) 2

21. Let $U_n = \frac{|n}{|n+2}$ where $n \in N$. If $S_n = \sum_{n=1}^{n} U_n$ then $\lim_{n\to\infty} S_n$ is equal to

(A) 2 (B) 1
(C) $\frac{1}{2}$ (D) none exists

Space for rough work

22. Given $z = \left(1+i\sqrt{3}\right)^{100}$ then $\frac{Re(z)}{Im(z)}$ is equal to

(A) 2^{100} (B) 2^{50}

(C) $\frac{1}{\sqrt{3}}$ (D) $\sqrt{3}$

23. If $|z - 1| \le 2$ and $|wz - 1 - w^2| = a$ (where w n a cube root of unity) then complete set of values of a is

(A) $0 \le a \le 2$ (B) $\frac{1}{2} \le a \le \frac{\sqrt{3}}{2}$

(C) $\frac{\sqrt{3}}{2} - \frac{1}{2} \le a \le \frac{1}{2} + \frac{\sqrt{3}}{2}$ (D) $0 \le a \le 4$

24. The number of positive integral solution of $x^4 - y^4 = 3789108$ is

(A) 0 (B) 1

(C) 2 (D) 4

25. The total number of ways of selecting two numbers from a set {1, 2, 3, 4 ….. 3n} so that their sum is divisible by 3 is equal to

(A) $\frac{2n^2 - n}{2}$ (B) $\frac{3n^2 - n}{2}$

(C) $2n^2 - n$ (D) $3n^2 - n$

26. The sum of the coefficients of even power of x in the expansion of $(1 + x + x^2 + x^3)^5$ is

(A) 256 (B) 128

(C) 512 (D) 64

27. Consider $f(x) = x^3 + ax^2 + bx + c$, parameters a, b, c are chosen respectively by throwing a dice three times. Then probability that f(x) is an increasing functions is

(A) $\frac{5}{36}$ (B) $\frac{8}{36}$

(C) $\frac{4}{9}$ (D) $\frac{1}{3}$

Space for rough work

28. If a, b, c $\in R^+$ then the minimum value of $a(b^2 + c^2) + b(c^2 + a^2) + c(a^2 + b^2)$ is equal to
(A) abc (B) 2abc
(C) 3abc (D) 6abc

29. $\sum_{r=1}^{n} \tan^{-1}\left(\frac{2^{r-1}}{1+2^{2r-1}}\right)$ is equal to

(A) $\tan^{-1}(2^n)$ (B) $\tan^{-1}(2^n) - \frac{\pi}{4}$

(C) $\tan^{-1}(2^{n+1})$ (D) $\tan^{-1}(2^{n+1}) - \frac{\pi}{4}$

30. In ΔABC, $a^2 + c^2 = 2002b^2$ then $\frac{\cot A + \cot C}{\cot B}$ is equal to

(A) $\frac{1}{2001}$ (B) $\frac{2}{2001}$

(C) $\frac{3}{2001}$ (D) $\frac{4}{2001}$

Space for rough work